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Mathematics in AI – Math 4312

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Investigating Differentiability Using R Shiny

For this project, I used the R Shiny app that I made earlier in the semester about differentiability. The app lets you choose different functions, change some of the values, and see how the graph and calculations change. This time, I wanted to look more at the numbers and see how much of a difference changing those values actually makes.

When we first started learning about differentiability, I had a hard time understanding how a function could be continuous but not differentiable. I knew that differentiability had something to do with the slope, but it was harder to understand why some functions had a derivative at certain points and others didn't. Being able to see the graphs and the calculations together helped me understand it better.

For this report, I wanted to see what happens when I make the value of h smaller. In the app, h is the distance used to estimate the slope. I was curious if using a smaller number would make the estimates more accurate or if it would depend on the function I was testing.

I tested four functions in my app: x squared, absolute value, a vertical tangent function, and a piecewise function. I looked at the slope estimates from both sides and compared the results. I

also wanted to see why some functions could have two completely different slopes or why a function might not have a derivative even when the numbers looked similar.

Mathematical Formulation

To understand how the app works, I had to use the formula for finding a derivative. A derivative tells us the slope of a function at a certain point. The formula uses something called a difference quotient, which looks at how much the function changes between two points.

The formula is:

$$\frac{f(a+h)-f(a)}{h}$$

In this formula, a is the point I'm testing, and h is the small distance between the two points. The app uses this formula to estimate the slope instead of finding every derivative by hand.

The app also calculates the slope from the left and from the right. The left slope uses the point just before a, while the right slope uses the point just after a. If both sides get closer to the same finite number as h gets smaller, the function is differentiable at that point.

For the tangent slope estimate, my app takes the average of the left and right estimates. This helps give an estimate of the slope at that point, although it doesn't always mean the function is differentiable.

For example, with the function $f(x) = x^2$ at $a = 2$, the actual derivative is 4. When I used $h = 0.1$, the app gave me a left slope of 3.9 and a right slope of 4.1. The average was 4, which matched the actual derivative.

This is how the app uses numerical calculations to help explain differentiability. Instead of only looking at the graph, I can compare the numbers and see whether the slopes are getting closer together.

Methods and Results

For this part of the project, I went back into my R Shiny app and tested each of the four functions. I used a different testing point depending on the function. For x^2, I also changed the value of h to see how it affected the slope estimates.

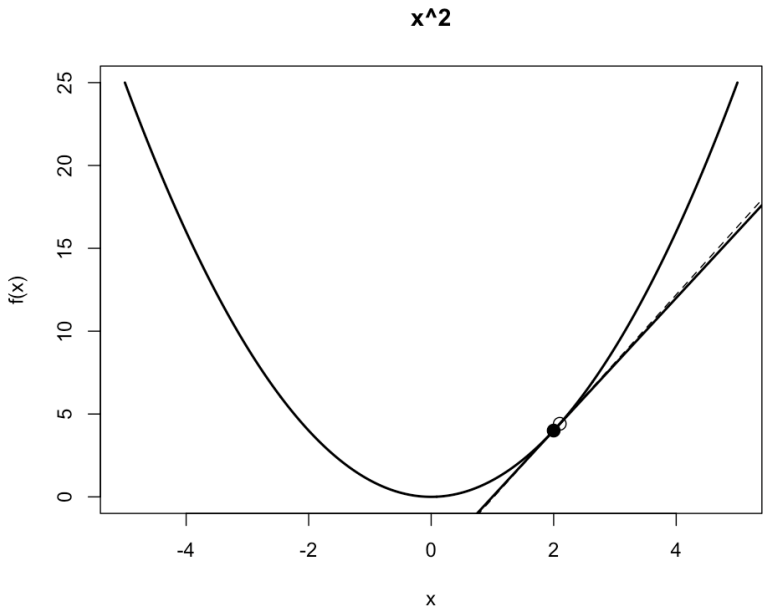
I started with the function $f(x) = x^2$ because it was the easiest one to compare with an actual derivative. I used $a = 2$ and first set $h = 0.1$. The app calculated a left slope of 3.9 and a right slope of 4.1. When I changed h to 0.01, the left slope changed to 3.99 and the right slope changed to 4.01.

Both estimates got closer to 4 when I made h smaller. Since the actual derivative of x^2 at $x = 2$ is 4, this showed me that using a smaller h gave a more accurate estimate in this example.

Next, I tested the absolute value function, $f(x) = |x|$, at $a = 0$ with $h = 0.1$. The left slope was -1 and the right slope was 1. Since the slopes were different, the app showed that the function was likely not differentiable at that point.

I also tested the vertical tangent function and the piecewise function at $a = 0$. These gave different results from x^2, which helped me see that not every function has a regular tangent slope at every point.

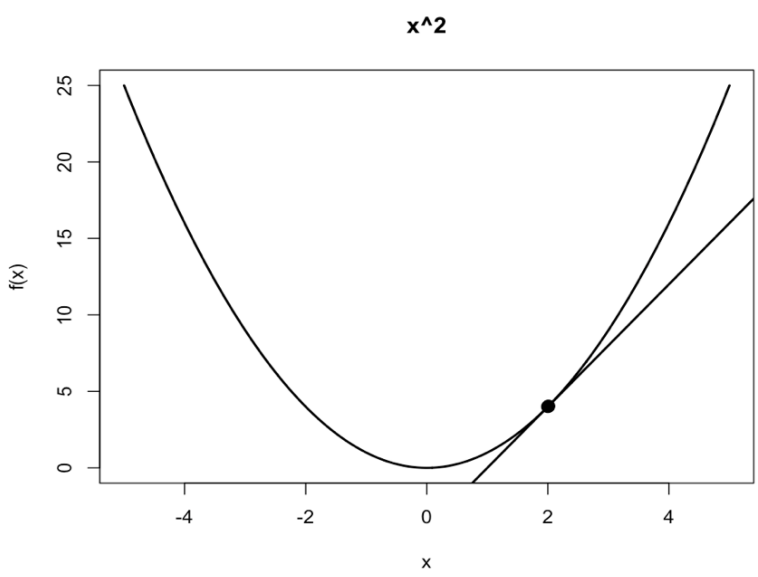
Function	Point (a)	h	Left slope	Right slope
x^2	2	0.1	3.9	4.1
x^2	2	0.01	3.99	4.01
$\|x\|$	0	0.1	-1	1
Vertical Tangent	0	0.1	4.6416	4.6416
Piecewise jump	0	0.1	21	1



Numerical Diagnostics

```
a = 2
h = 0.1
f(a) = 4
Difference quotient = 4.1
Left derivative estimate = 3.9
Right derivative estimate = 4.1
Tangent slope estimate = 4
Conclusion: Likely differentiable at a
```

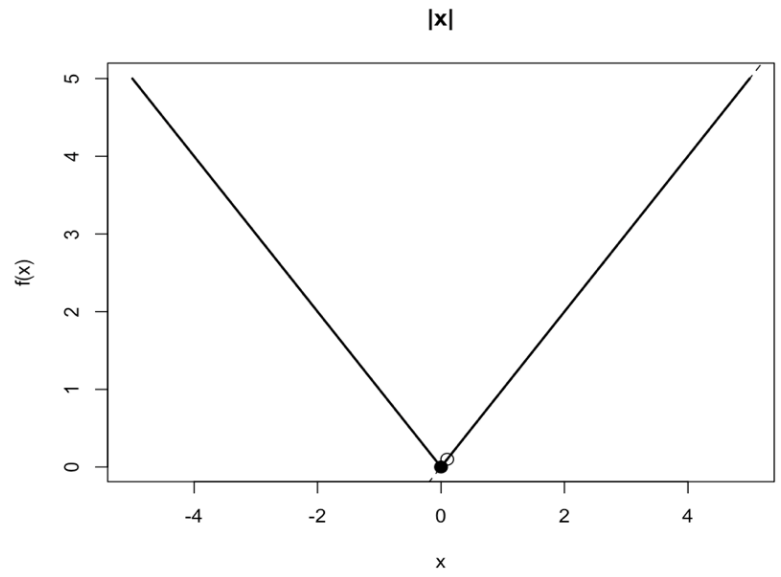
Figure 1. x^2 at $a = 2$, $h = 0.1$



Numerical Diagnostics

```
a = 2
h = 0.01
f(a) = 4
Difference quotient = 4.01
Left derivative estimate = 3.99
Right derivative estimate = 4.01
Tangent slope estimate = 4
Conclusion: Likely differentiable at a
```

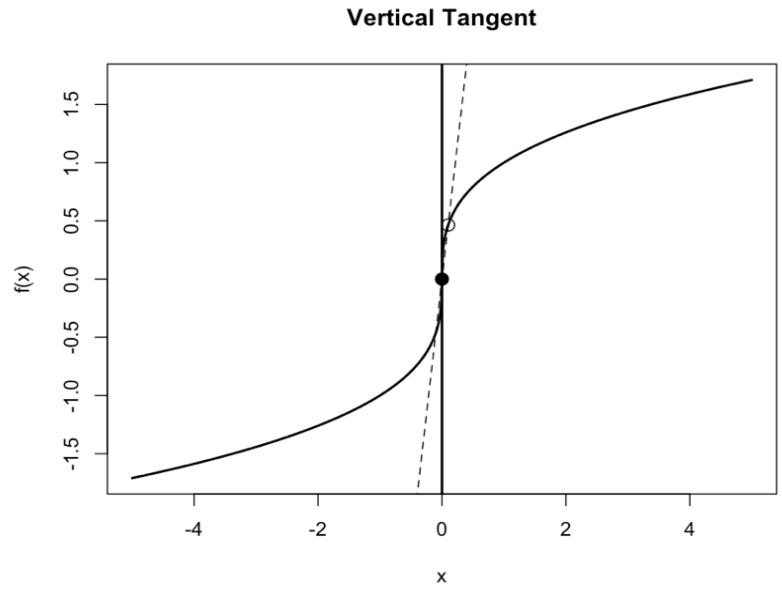
Figure 2. x^2 at $a = 2$, $h = 0.01$



Numerical Diagnostics

```
a = 0
h = 0.1
f(a) = 0
Difference quotient = 1
Left derivative estimate = -1
Right derivative estimate = 1
Tangent slope estimate = Does not exist
Conclusion: Likely not differentiable at a
```

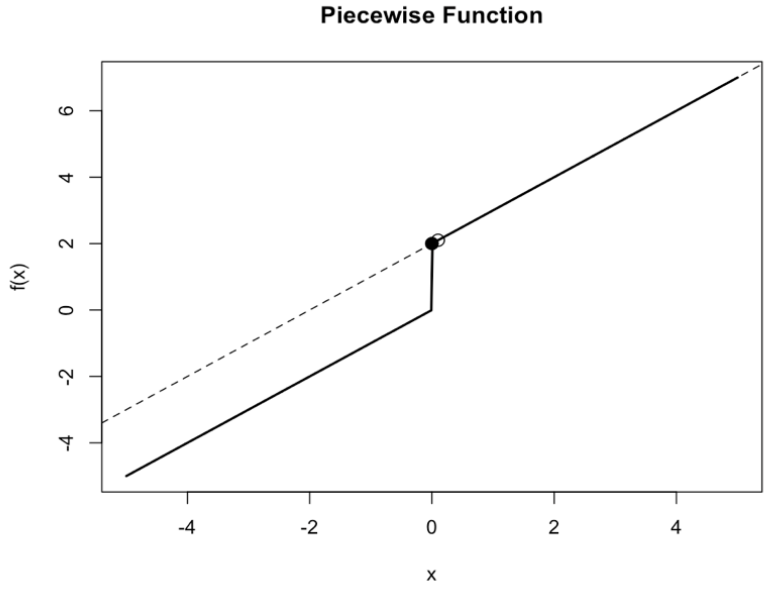
Figure 3. $|x|$ at $a = 0$, $h = 0.1$



Numerical Diagnostics

```
a = 0
h = 0.1
f(a) = 0
Difference quotient = 4.6416
Left derivative estimate = 4.6416
Right derivative estimate = 4.6416
Tangent slope estimate = Vertical
Conclusion: Likely not differentiable at a
```

Figure 4. Vertical tangent at a = 0, h = 0.1



Numerical Diagnostics

```
a = 0
h = 0.1
f(a) = 2
Difference quotient = 1
Left derivative estimate = 21
Right derivative estimate = 1
Tangent slope estimate = Does not exist
Conclusion: Likely not differentiable at a
```

Figure 5. Piecewise function at a = 0, h = 0.1

Understanding The Results

When I looked at the numbers in my table, the biggest thing I noticed was how much closer the slopes got when I made h smaller for x^2. With h at 0.1, I got 3.9 and 4.1, but when I changed it to 0.01, I got 3.99 and 4.01. I knew the actual slope was supposed to be 4, so it was interesting to see how close the app could get just by changing that one value.

The absolute value function was pretty straightforward once I saw the numbers. At $x = 0$, one side gave me -1 and the other gave me 1. Looking at the graph, it made more sense because there's a corner right there. The slopes are going in different directions, so there isn't one slope at that point.

The vertical tangent function was the one that threw me off a little. Both sides gave me 4.6416, which made me wonder why the app said it wasn't differentiable. The difference is that those numbers don't stay around 4.6416 when h gets smaller. They keep increasing instead of getting closer to one finite slope.

For the piecewise function, I got 21 on the left and 1 on the right. That's a pretty big difference, but when I looked at the graph, I could see that the function jumps at $x = 0$. The graph isn't even connected at that point, so it makes sense that there wouldn't be a derivative there.

I think the biggest thing I got out of testing these functions was realizing that the app can give me numbers for the slopes even when a derivative doesn't actually exist. I still have to look at what those numbers are doing and what the graph looks like instead of just assuming the calculation means there's a derivative.

Assumptions and Limitations

One thing I had to keep in mind while using my app was that the slopes were only estimates. The app uses the value of h to calculate them, so the results can change depending on what number I choose. Even though making h smaller helped with x^2, that doesn't mean it will always give me a perfect answer for every function.

Another limitation is that my app only has four functions to choose from. I picked those because they show different examples of differentiability, but there are obviously a lot more functions that I could have tested.

The app also uses calculations to decide whether a function is likely differentiable. It compares the slopes from both sides, but that isn't the same as proving it mathematically. For some of the functions, I also had to include special conditions in the code so the app would recognize points where the derivative doesn't exist.

I also noticed that the value of h can make some results look confusing. For example, the vertical tangent function gave me matching slopes, but that didn't mean the derivative existed. If I had only looked at those two numbers without understanding the function, I could have gotten the wrong idea.

I think the app works well for showing how differentiability works, especially when I can see the graph and the numbers together. But I wouldn't rely on it alone to decide whether every function is differentiable.

Based on the feedback I received on my original project, one area I could improve is how the app determines whether a function is differentiable. Right now, it compares the left and right slope estimates, but it doesn't test several smaller values of h to see whether they approach the same finite number. I would like to improve this by having the app check multiple values of h and include an “inconclusive” result when there isn't enough information to make a decision. I think these changes would make the app more accurate and reliable.